

EngD thesis examination summary

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Thesis: From FIDS to FIAS: A Behavioural Science Programme on Failure-Induced Abandonment in Gamified Fall Risk Assessment, Toward an Adaptive-Learning Framework.

Submission reviewed: 13082026v4_Full_version_V37_REVIEW_EngD_Thesis_Submission_Draft.pdf

Review date: 5 September 2026 | 193 PDF pages

Provisional judgement: major revision before submission

This is a worthwhile, practice-led engineering doctorate project with a compelling field problem, an implemented game-based intervention, and a potentially useful behavioural design framework. However, I would not recommend acceptance or minor corrections on the strength of V37. The central argument repeatedly exceeds the evidence, and material questions about consent, chronology, outcome ascertainment and source records remain unresolved.

My examiner-style position would be to require substantial revision and inspect the supporting records before deciding whether the work meets the doctoral standard. This is an academic judgement, not an official PolyU examination category or prediction of your examination outcome.

Dimension	Assessment
Industrial relevance	Strong: underused assessment equipment is a concrete deployment problem.
Original contribution	Promising: a context-specific failure/return account translated into design requirements; its novelty needs a sharper comparison with prior work.
Empirical evidence	Useful descriptive observations; causal and psychological mechanism claims remain unestablished.
Engineering rigour	A credible architecture is described, but the deployed builds, measurements and AI operation need an auditable record.
Submission readiness	Not ready: substantive evidence issues and unresolved review-copy material remain.

The strongest defensible thesis

The programme documents low engagement in one residential deployment, reports substantially more return in a different home-play context, and uses those observations to propose a testable design framework. That contribution can be valuable without claiming to have falsified gamification theory or discovered the unique cause of abandonment.

Review scope: Examination of the supplied thesis, including its appendices; selected visual checks; arithmetic checks from printed counts; and targeted primary-source checks. The referenced raw workbooks, code, approval memo and consent records were not supplied and were not independently audited. All thesis page references below use printed page numbers; add 18 for the PDF viewer page, except the front matter.

What is strong, and what the evidence supports

Strengths worth preserving

A clear EngD problem. The unused FRA station provides a convincing connection between engineering design and real service delivery. Your practitioner access is an asset when accompanied by reflexivity and an audit trail (Chapters 1 and 3).

A useful focus on non-return. Studying people who stop using a system, rather than reporting only satisfied completers, is valuable. The distinction between recorded performance, satisfaction and future use deserves to remain central (pp. 75-85).

Engineering translation. The sensing/game architecture, QFD mapping and three design principles connect observations to actionable system changes (pp. 59-74, 79-81, 105-109). The most credible originality is in this particular integration and its application.

Several responsible qualifications. V37 explicitly withdraws unsupported survival and motivational-orientation estimates, acknowledges outcome contamination in the FIAS score, and says the F4 interviews were not conducted. These disclosures are appropriate. They now need to govern every conclusion consistently.

Evidence in V37	Permissible conclusion / principal limit
Stage 1: N=200; 39% prior use; approximately 74% adverse first experience and 82% non-return (p. 53).	A retrospective problem signal. Conditional denominators and item mappings need reconciliation; misunderstanding or non-acceptance is not automatically task failure.
Stage 2: 21/30 completed no more than three cycles; reported return 0/21 (p. 76).	Low session participation and reported non-return within the window. Cycle totals alone do not establish the timing, cause or permanence of abandonment.
Stage 2: five of eight single-session users recorded an index increase and did not return (pp. 77, 81-82).	Performance-score direction and return can diverge. This does not rule out fatigue, boredom, measurement error or lack of perceived benefit.
Stage 3: 15/23 initial dropouts returned; in ages 65-80, 10/16 returned (pp. 89-90).	A promising return pattern under a different deployment package. These were different people, apparatus, settings and observation conditions.
Adaptive-learning framework and dynamic states (Chapters 7-9).	An evidence-informed design proposal. The complete package has no reported controlled effectiveness evaluation in this PDF.

Arithmetic spot-check: The reported Wilson intervals for 21/30, 0/21, 15/23, 10/16, 6/41, 86/120 and 23/41 reproduce to one decimal place. The satisfaction comparison (9/9 versus 11/21) gives Fisher $p=0.01340$, consistent with the thesis. Paired-score tests and individual classifications cannot be reproduced without the raw records.

Priority 1: resolve provenance and consent

1. Ethics chronology needs documentary resolution

Stage 1 is dated July-August 2024, Stage 3 March-April 2026, and approval August 2026; Stage 2 still lacks exact dates (pp. 42-44). Section 3.4 says Stage-2 participants were not told that play was research; written consent was obtained retrospectively from sustained participants, while dropouts did not sign. Yet the analysis includes all 30 participants.

Required action: Present the dated approval memo, approved protocol/version, any earlier approval or exemption, the authorised scope of incomplete disclosure, debriefing arrangements, and the documented basis for using each group's records, including non-consenting dropouts. Seek a written institutional determination where coverage is unclear. A statement that a later application described earlier activities does not, by itself, resolve their status. The university's public guidance requires ethical review [1]; only the actual institutional record can settle this case. This report does not infer misconduct.

2. The archive is asserted to be complete but is not evidenced here

Section 8.9 (p. 133) says complete instruments, coding and reliability are bound. Appendix B (p. 163) expressly says the codebook is not reproduced and is future work. The opening checklist still requests source survey items, scales, raw responses, AI details and participant compilation. Appendix G lists filenames without the actual datasets or a usable examiner access procedure in this PDF (pp. 171-172).

Required action: Create a source-to-result manifest: source file, collection date, anonymised record IDs, cleaning/derivation script, output table, access location and verification status. Every claimed deposited item must actually be accessible. Preserve raw values and an amendment log; withdraw or qualify results that cannot be reconstructed. Synthetic pipeline-test records are correctly labelled in G.8 and must remain wholly separate from study evidence.

3. Repair contradictory archive references and dates

G.8 is cited for a vendor brochure, development videos and an expert guide (pp. 49, 66, 71, 173), but its actual entry is a synthetic workbook. G.4 lists consented deployment photographs, although Figure 5.1A says no participant play imagery exists. Resolve each reference to one exact artefact; distinguish equipment photographs, control demonstrations and study participants.

The title page says June 2026; the draft contains August consultations and an October 2026 completion entry. Stage 3B has no results and pending dates. Freeze one evidence cut-off and label all later work as planned. The 2023-2024 design chronology also needs to distinguish early prototypes from later Claude Code development: Anthropic announced Claude Code on 24 February 2025 [2]. This is a chronology question, not proof that the earlier prototype did not exist.

Why this comes first: Wording improvements cannot compensate for unverified source records or unresolved permission to analyse them. If the institutional determination limits usable records, re-run the analysis and reassess the contribution on the permitted evidence.

Priority 2: make outcomes and measurements auditable

4. Reconstruct the participant journey, not just totals

Stage 2 defines a cycle as an attended session under a five-attempt rule (p. 75). Its archive lists cycle counts but no timestamps or explicit first-failure variable. Nevertheless, the thesis claims a strong first-failure association and “faster” dropout (p. 84). Report the actual failure definition, source and failure-by-disengagement table. Without those records, retain a low-engagement description and remove causal or temporal wording.

For Stage 3 (p. 88), initial dropout includes “no second session within the study window”, while return is a subsequent session after dropout. The timing needs a landmark: for example, an initial ascertainment interval followed by a separately defined return interval. Reconstruct the definition actually used; do not retrospectively invent a favourable rule. Distinguish within-session retry, session completion, later attendance and end-window non-return.

Document invitations, game starts, returned logs, questionnaire returns, missing files, duplicate IDs, contact/reminder history and usable follow-up. Establish whether N=120 represents all enrolled starters or only people who returned data. A missing emailed log may mean missing observation, not non-use; conversely, analysing only respondents can inflate retention. State the limits if invitation denominators are unknowable.

5. Clarify what the FRA index actually measures

The thesis alternates between a single-session change for the F4 cases and a pre/post eight-week measurement for all participants (pp. 75-77, 81-84). Specify each measurement’s time relative to gameplay and exit, particularly for participants said never to have returned. Explain how complete post-values were obtained for them.

Identify the exact vendor score, units, direction, acquisition protocol, calculation and interpretation. Sharing a pressure signal does not establish that unrestricted gameplay produces the same validated measurement as the standard assessment. Chapters 4 and 7 recognise this, but Chapter 5 also asserts a silent clinical pipeline and uncompromised dual use (pp. 62-64). Park et al. support association with established performance tests under their study protocol [3], not automatic validity of this game-derived implementation.

6. Reconcile instruments and denominators

Appendix A item 3 combines understanding with acceptance. Report exact counts for prior users, negative answers and non-return, including their cross-tabulation; p. 38 cannot say every percentage uses 200 when some are conditional. Appendix C does not reproduce the administered after-game form, while Chapter 5 points to questionnaires in Appendix B, which instead contains a coding summary. The Stage-3 five-point scale must be reconciled with the cover checklist’s unresolved five-versus-seven-point question.

Keep the FIAS composite descriptive or remove it until its scoring is reproducible. Provide the verification failure reason and actual sensitivity results for the unverified Stage-2 record. Treat satisfaction as a contemporaneous/post-use association, not an established cause of staying.

Priority 3: rebuild the argument at the right strength

7. H0 is an overstrong theoretical expectation

The central H0 combines acceptance, sustained use and absence of lasting abandonment (p. 19). No numerical null, rejection threshold or measured TAM/STAM compliance is supplied. A designed feature is not proof that users experienced the intended construct. The lack of a comparator also prevents a test of improved acceptance. Section 3.5 correctly calls H0 reconstructed during writing (p. 45), whereas other sections present its rejection as prospective theory falsification.

Revision: Present a theory-informed design expectation that the deployment did not fulfil. Retain explicit directional research questions and identifiable statistical tests. Do not say that satisfying a design checklist eliminates implementation quality as an explanation (p. 59). An unsuccessful deployment cannot falsify the entire acceptance/gamification literature.

8. Stage 3 cannot establish a coupling “scope law”

The comparison changes residential status, health profile, recruitment, supervision, physical effort, input device, platform and clinical procedures. The 65-80 band is the closest available age comparison, not a matched control. Different dropout definitions also select different subsets for the return-rate comparison. The age gradient within the decoupled condition cannot estimate an age-by-clinical-frame interaction because clinical framing does not vary within that cohort.

Replace “attribution experiment”, “positive control” and the universal boundary formulation with a non-equivalent-cohort comparison that motivates a framing hypothesis. In particular, the cautious opening of section 6.7 contradicts its subsequent general boundary claim (p. 93); the same problem recurs on pp. 119, 139-140 and 151-152.

9. Take simpler explanations seriously

The score/return divergence is interesting but not logically paradoxical: a person can improve slightly and still find an activity unpleasant, tiring, unnecessary or inconvenient. Section 8.5.9 cannot eliminate ordinary attrition using timing that was not retained, nor assume boredom is incompatible with improvement. Section 4.1A also describes real consequences of assessment: prescribed training and restrictions on independent walking (p. 52). Avoidance of those consequences is a strong rival to an unmeasured identity mechanism.

FIAS should remain a provisional failure-associated non-return construct. Do not infer loss aversion, stereotype threat or mental accounting from non-return alone. Compare the proposed increment against continuance theory, self-efficacy, information avoidance, autonomy/reactance and existing abandonment accounts. Bhattacharjee’s established post-adoption continuance model is a particularly relevant omission [4].

10. Adaptive learning is a justified proposal, not the only solution

The 29.3-point retention/intention difference compares distinct outcomes; it does not diagnose missing adaptive learning or show that half of retained users lack intention without a joint table. The claim that the framework is the only safe response (p. 119) contradicts p. 13. Its originality must be the specific architecture and testable propositions, not the discovery of adaptive learning itself.

Priority 4: strengthen the engineering contribution

11. Separate deployed, demonstrated and proposed systems

Supply one version-controlled matrix for the 2024 control demonstration, Stage-2 deployment, Stage-3 HTML port, successor screenshots, dynamic-state prototype and Stage 3B. Record dates, code version/hash, hardware, controls, rules, clinical acquisition, runtime AI, logging, and the participants who used each build.

Table 7.2 says Stage 2 and Stage 3 use the same pressure-panel hardware (p. 107), contradicting Chapter 6 and Appendix E. The final synthesis says the deployed system was deterministic (p. 152), whereas Chapter 5 describes Stage-2 LLM calls. Appendix E clearly states Stage 3 was rule-based. Specify the actual distinction, including whether Stage-2 calls occurred and what evidence survives. The full endpoint configuration is not actually printed in G.7.

Likewise, distinguish logging capability from retained logs. Appendix E lists timestamps and AI-call telemetry (pp. 167-168), while the analysis archive lacks timestamps. That can reflect data loss or a reduced export, but the thesis must state which. Resolve layer-number differences between the original stack and the dynamic-state specification.

12. Make requirements verifiable

The QFD matrix (pp. 79-81) is a useful start, but its user-need statements are researcher interpretations, not demonstrated verbatim user requirements. No numerical importance weights or prioritisation calculation is shown, and the coin-reward column mentioned in prose is absent. Either complete an auditable QFD method or call this a qualitative requirements mapping. Retention among completers does not isolate a coin-reward effect.

Engineering question	Evidence that would strengthen the submission
How are observations translated into design?	A compact requirements trace: observation → proposed need → feature → acceptance criterion → verification result.
Did the adaptive controller actually act?	Representative authentic event traces, runtime call/fallback records, parameter bounds, failures and manual overrides; state missing records explicitly.
Why a five-minute update?	Relate update timing to actual run/session duration and show whether short-lived users ever received an update.
What does AI add?	Separate authoring assistance from runtime adaptation. Compare against a transparent rule baseline where results exist; otherwise leave added value untested.
Can industry use the output?	A bounded implementation specification with staffing/training needs, maintenance, accessibility and rough resource estimates, with assumptions labelled.

Stage 3B's adaptive/static × zero-failure/failure-possible design can test those factors. Without a clinical-framing factor it cannot itself isolate the clinical-coupling claim. If included, provide the dated registration, allocation, primary contrast, sample-size rationale, analysis plan, amendments and all results; otherwise keep it outside the completed empirical contribution (pp. 129-132).

Chapter comments and revision sequence

Chapter	Examiner comment / principal revision
1 · Introduction	Keep the industrial opening. Bound sector-wide claims; align aims with observational evidence. Use one stable set of research-question labels.
2 · Literature	Retain the integrated account, but replace the forced H0 narrative with critical comparison. Add post-adoption and avoidance alternatives; substantiate absolute novelty claims.
3 · Methods	Resolve dates and ethical status first. Provide recruitment flow, actual instruments, operational outcomes, hypothesis chronology and action-research decision records.
4 · Stage 1	Reconcile exact conditional counts and survey wording. Remove unsupported claims that all acceptance conditions were met or that excluded residents were unlikely to differ.
5 · Stage 2	Recover measurement/failure provenance and build history. Keep descriptive findings; remove unsupported “faster dropout”, mechanism exclusion and unverified score interpretations.
6 · Stage 3	Separate session retry from return. Reconcile the denominator and follow-up; treat the comparison as a package/context association. Move future model comparisons out of results.
7 · Framework	Make this the engineering centre: requirements, architecture, trade-offs and verification. Remove claims that three principles are empirically exhaustive or uniquely necessary.
8 · Discussion	Keep plausible alternatives genuinely open. Expert accounts inform interpretation; they do not validate mechanisms. Complete member checks and distinguish the consultation chronology.
9 · Conclusion	Answer each research question once. Remove “eliminated”, “scope law” and proven-restoration language. Keep only completed results in the evidential chain.
Appendices / layout	Provide actual instruments and usable archive access. Resolve G.8 collisions, photo descriptions, stale contents entries, missing section references and review highlights.

Recommended order of work

First: Resolve institutional coverage and establish the authoritative source archive. This determines which results can remain.

Second: Rebuild the participant/outcome tables, measurement provenance and software history; reproduce the analyses from permissible records.

Third: Rewrite the abstract, H0 architecture and conclusions to match those results; sharpen the contribution against prior literature.

Fourth: Complete the requirements-to-verification account and appendices, then refresh contents, figures, references and pagination.

I would not automatically demand a new large trial for a bounded exploratory EngD. A defensible design contribution may be possible with the present programme if its evidence is valid and sufficiently complete. However, a causal claim about clinical framing, or a claim that the framework prevents abandonment, requires new controlled evidence. Future-work promises cannot substitute for the contribution being examined.

Viva preparation and final assessment

Questions I would ask you in the examination

1. What original contribution remains if FIAS is not a validated syndrome?
2. Where is first failure independently recorded, and how does it differ from dropout?
3. Why is observed non-return evidence for identity appraisal rather than burden or avoidance of real restrictions?
4. How would you distinguish clinical framing from input modality and supervision?
5. What exact score was measured, when, and how were post-values obtained for single-session exits?
6. Which framework components were deployed, and what evidence shows they worked?
7. What did the LLM do that a rule controller could not, and was that tested?
8. Which records establish consent/approval coverage and reproduce each headline result?

My concluding opinion

Sidney, I see a credible EngD opportunity in your engineering response to a difficult human adoption problem. I would defend the value of that problem and the usefulness of your design thinking. I would challenge the claim that the programme has already identified the mechanism, excluded conventional explanations, or restored clinically useful engagement through the complete framework.

The revision should increase precision, not rhetorical force. Keep the observed phenomenon, make the record independently inspectable, and distinguish your proposed explanation from your implemented and verified contribution. If those conditions are met, the thesis will be much easier to defend. If the central records cannot be reconciled, the acceptable scope and doctoral sufficiency will need to be reconsidered with your supervisor.

Targeted source checks used in this review

These checks supplement the supplied thesis; they are not a complete reference audit. Links accessed 5 September 2026.

[1] The Hong Kong Polytechnic University. **Human Subjects Ethics Application Review System**. Establishes the institutional ethical-review requirement; it does not resolve this programme's individual approval status.
[PolyU HSEARS guidance](#)

[2] Anthropic (24 February 2025). **Claude 3.7 Sonnet and Claude Code**. Primary announcement dates Claude Code's limited research preview.
[Anthropic launch announcement](#)

[3] Park et al. (2019). **Introduction of Fall Risk Assessment (FRA) System and Cross-Sectional Validation Among Community-Dwelling Older Adults**. *Annals of Rehabilitation Medicine*, 43, 87-95. The abstract supports association with performance tests, not this thesis's game-derived protocol.
[doi:10.5535/arm.2019.43.1.87](https://doi.org/10.5535/arm.2019.43.1.87)

[4] Bhattacharjee (2001). **Understanding Information Systems Continuance: An Expectation-Confirmation Model**. *MIS Quarterly*, 25(3), 351-370. Relevant comparator for the post-adoption gap.
[Association for Information Systems article record](#)

[5] An, Cheung & Willoughby (2024). **A gamification approach for enhancing older adults' technology adoption and knowledge transfer**. The PolyU record confirms article 123456 and its DOI; that number is genuine, not a placeholder. Its abstract reports a one-group pre/post design and improvements in knowledge and adoption intention; it supplies no universal retention threshold for the thesis's H0.
[PolyU Scholars Hub publication record](#)